Self-Supervised and Semi-Supervised Hybrid Learning

Goal: Build a good classification model using **mostly unlabeled data** + a little labeled data.

Examples:

- Self-supervised pre-training (contrastive like SimCLR/MoCo, or reconstruction) on unlabeled skin/thermal/leaf images → then fine-tune on few labels
- Semi-supervised (FixMatch, pseudo-labeling, self-training) to use lots of unlabeled + small, labeled set for skin/breast cancer/plant disease
- Good for medical (skin, thermography, H&E tissue, endoscopy) or agri (jute/tomato/multi-fruit) where labeling is hard/expensive

Phase 0 – Preparation (Before Starting)

□ Decide your main approach today

- Pure Self-Supervised → pre-train on ALL unlabeled data → fine-tune on labeled
- Semi-Supervised → use small labeled + large unlabeled together (e.g., pseudo-labeling / consistency regularization)
- Combined: Self-supervised pre-train first → then semi-supervised fine-tune

Chosen type today: (e.g., Self-Supervised + Fine-Tune / Semi-Supervised FixMatch-style)

□ Dataset structure

- Unlabeled folder (big!): thousands of images (no labels needed)
- Labeled folder (small): train/val/test with labels Examples:

```
├── unlabeled (all skin/leaf/thermal images)
├── labeled_train
├── labeled_val
└── labeled_test
```

□ Decide ratio:labeled vs unlabeled

Labeled: _____ images (e.g., 5-20% of total)

Unlabeled: ____ images (e.g., 80-95%)

□ Choose pre-training task (for self-supervised) Common:

- Contrastive (SimCLR, MoCo): augment same image twice → make similar
- Masked image modeling (MAE): reconstruct masked parts
- Rotation prediction / jigsaw / colorization

Choose backbone: _____ (e.g., ResNet50, ViT-small, EfficientNet)

Phase 1 – Data Preparation Checklist

□ All images resized (224×224 or 256×256)

□ Normalized (ImageNet stats usually)

□ Strong augmentations for self/semi (very important!)

- Random crop, flip, color jitter, blur, cutout, mixup, etc.
- Use two views per image for contrastive

□ Labeled split ready: train/val/test

□ Unlabeled folder ready (no labels needed)
□ Class imbalance noted (common in disease data) → plan: focal loss / weighted sampling later
□ Numbers written: Total unlabeled: ___ Labeled train: ____ Val: ____ Test: ____

Phase 2 – Choose & Prepare ONE Experiment

Experiment A – Pure Self-Supervised pre-train (e.g., SimCLR/MoCo on unlabeled) → fine-tune on labeled
Experiment B – Semi-Supervised only (e.g., pseudo-labeling / self-training / FixMatch on labeled + unlabeled)
Experiment C – Combined: Self-supervised pre-train → semi-supervised fine-tune
Experiment D – Simple baseline: Supervised only on labeled (to compare improvement) or others.

Phase 3 – Model Building Steps (Paper Checklist)
□ Step 3.1 – Load pre-trained backbone (or from scratch for SSL)
□ Step 3.2 – For self-supervised: add projection head (small MLP for contrastive)
□ Step 3.3 – Prepare augmentations pipeline (strong for SSL/SSL)
□ Step 3.4 – For semi-supervised: decide consistency method (e.g., mean-teacher, pseudo-label threshold)
□ Step 3.5 – Loss:
- Self-supervised → NT-Xent / MSE reconstruction
- Semi-supervised → supervised CE + unsupervised consistency loss

□ Step 3.6 – Optimizer: AdamW / LARS (good for SSL)
□ Step 3.7 – Learning rate: higher for pre-train (e.g., 0.03–0.1), lower for fine-tune (1e-4)
□ Step 3.8 – Batch size: large if possible (256–1024 for contrastive)

Phase 4 – Training Workflow (Two-Stage Usually)
Stage 1: Pre-training (Self-Supervised – long, on unlabeled only)
1. Train for many epochs (100–1000, depend on data)
2. Monitor: loss decreases, no labels so no accuracy yet
3. Save encoder/backbone weights (remove projection head)

Stage 2: Fine-tuning / Semi-Supervised
1. Load pre-trained encoder
2. Add classification head (linear / small MLP) → match num_classes
3. For pure semi: train with labeled + unlabeled (pseudo-label or consistency)
4. For each epoch:
   □ Train pass (labeled loss + unsupervised if semi)
   □ Validate on labeled val → accuracy / F1
   □ Save best (early stopping)
5. Finish → keep best model

Phase 5 – Specific Actions by Type
Self-Supervised Pre-train
□ Use large batch / strong aug. → learn good features

□ Freeze encoder first in fine-tune, then unfreeze

Semi-Supervised

□ Start with high confidence threshold for pseudo-labels (e.g., 0.95)

□ Gradually lower threshold or add more unlabeled

□ Monitor over-fitting on small labeled set

Phase 6 – Final Evaluation (Must Do)

□ Test only once on **held out labeled test set**

□ Calculate & note (focus on F1 because imbalanced diseases):

Metric	**Value**
Accuracy	____ %
Weighted F1	____
Macro F1	____
Per-class F1	list them

□ Compare with:

- Supervised baseline (only labeled data)
- How much gain from unlabeled? (+___ % F1)